

SCH4U Lab: pH of Salts and K_a

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Introduction

This lab explores the pH of salt solutions, the acid dissociation of weak acids, and the properties of buffer solutions. When a neutralization occurs, a salt is formed. Neutral salts form when a strong acid and a strong base react. A non-neutral salt forms when a weak acid or a weak base is involved, it exhibits either acidic or basic properties due to hydrolysis reactions in water.

Weak acids/bases only partially dissociate in aqueous solution. The strength of a weak acid or bases is quantified by its acid dissociation constant, K_a or K_b respectively. By measuring the pH of buffer solutions containing a weak acid/weak base and its conjugate base/acid, we can determine the K_a/K_b value. A buffer solution resists changes in pH when small amounts of acid or base are added, making it useful in many chemical and biological applications.

This lab consists of three parts. First, the pH of different salt solutions will be measured using a pH probe. Second, buffer solutions will be prepared by combining a weak acid with its conjugate base, and their pH will be recorded to calculate K_a values. Finally, the buffer's ability to resist pH changes will be tested by adding a strong base and comparing the pH change to that of pure water treated the same way.

If solutions contact skin, rinse immediately as they may cause skin irritation and handle glassware with care.

Procedure

Part 1: pH of Salt Solutions

<u>Materials/Equipment</u>	<u>Actions Taken</u>
Beaker (100 mL)	Prepare clean beaker
Graduated Cylinder	Measure and add 20.0 mL of 0.100 M NaCl
pH Probe	Calibrate probe and measure pH
	Record pH
Deionized Water	Repeat for Na_2CO_3 and NaHSO_3

Part 2: K_a of a Weak Acid

<u>Materials/Equipment</u>	<u>Actions Taken</u>
Beaker (250 mL)	Prepare clean beaker
Graduated Cylinder	Measure and add 20.0 mL of 0.1 M NaHSO_3 Measure and add 20.0 mL of 0.100 M Na_2SO_3
Glass Rod	Stir to mix
pH Probe	Measure and record pH Rinse probe Repeat with CH_3COOH and NaCH_3COO solutions

Part 3: Buffer Solution Testing

<u>Materials/Equipment</u>	<u>Actions Taken</u>
Buffer Solution (from Part 2)	Prepare 40.0 mL of $\text{NaHSO}_3/\text{Na}_2\text{SO}_3$ buffer
Graduated Cylinder	Add 10.0 mL of NaOH to buffer
pH Probe	Measure and record pH
Beaker	Prepare 40 mL of deionized water
Graduated Cylinder	Add 10.0 mL of NaOH to water
pH Probe	Measure and record pH

Observations

Table 1: pH Measurements of Salt Solutions

Solution	pH
0.100 M NaCl	5.6
0.100 M Na ₂ CO ₃	9.9
0.100 M NaHSO ₃	4.3

Table 2: pH of Buffer Solutions

Buffer	Acid	pH
NaHSO ₃ /Na ₂ SO ₃	HSO ₃ ⁻	6.5
NaCH ₃ COO/CH ₃ COOH	CH ₃ COOH	6.6

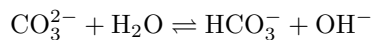
Table 3: pH Changes Upon Addition of NaOH

Solution	Initial pH	After 10 mL NaOH	pH Change
Buffer (HSO ₃ ⁻ /SO ₃ ²⁻)	6.5	6.9	0.4
Deionized Water	7.0	11.0	4.0

Analysis

Part 1: Predicted pH of Salt Solutions

Na₂CO₃ pH Calculation



Calculate the K_b value of CO_3^{2-} ($K_{a2} = 4.7 \times 10^{-11}$):

$$\begin{aligned} K_b &= \frac{K_w}{K_{a2}} \\ &= \frac{1.0 \times 10^{-14}}{4.7 \times 10^{-11}} \\ &= 2.127659 \times 10^{-4} \\ &= 2.1 \times 10^{-4} \end{aligned}$$

	$[\text{CO}_3^{2-}]$	$[\text{HCO}_3^-]$	$[\text{OH}^-]$
I (mol/L)	0.100	0	0
C (mol/L)	$-x$	$+x$	$+x$
E (mol/L)	$0.100 - x$	x	x

$$K_b = \frac{[\text{HCO}_3^-][\text{OH}^-]}{[\text{CO}_3^{2-}]} = \frac{x^2}{0.100 - x} = 2.127659 \times 10^{-4}$$

$$x^2 + 2.127659 \times 10^{-4}x - 2.127659 \times 10^{-5} = 0$$

$$x = [\text{OH}^-] = 4.50749 \times 10^{-3} \text{ mol/L} \rightarrow 4.5 \times 10^{-3} \text{ mol/L}$$

$$\begin{aligned} \text{pOH} &= -\log(4.50749 \times 10^{-3}) \\ &= 2.34606 \\ &= 2.35 \end{aligned}$$

$$\begin{aligned} \text{pH} &= 14.00 - 2.34606 \\ &= 11.65394 \\ &= 11.65 \end{aligned}$$

$\therefore$ The predicted pH of Na₂CO₃ is 11.65 (experimental: 9.9)

NaHSO₃ pH Calculation

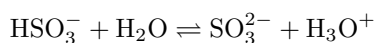
As an acid:

$$K_{a2} = 6.2 \times 10^{-8}$$

As a base:

$$\begin{aligned} K_b &= \frac{K_w}{K_{a1}} = \frac{1.0 \times 10^{-14}}{1.3 \times 10^{-2}} \\ &= 7.692307 \times 10^{-13} \\ &= 7.7 \times 10^{-13} \end{aligned}$$

Since $K_{a2} > K_b$, HSO_3^- acts as an acid:



	$[\text{HSO}_3^-]$	$[\text{SO}_3^{2-}]$	$[\text{H}_3\text{O}^+]$
I (mol/L)	0.100	0	0
C (mol/L)	$-x$	$+x$	$+x$
E (mol/L)	$0.100 - x$	x	x

Assume:

$$\frac{0.100}{6.2 \times 10^{-8}} = 1.6129 \times 10^6 > 1000$$

Assumption is valid.

Verify:

$$\frac{0.100-x}{0.100} \times 100\% = 99.9\% > 95\%$$

Verification successful.

$$K_{a2} = \frac{[\text{SO}_3^{2-}][\text{H}_3\text{O}^+]}{[\text{HSO}_3^-]} \approx \frac{x^2}{0.100} = 6.2 \times 10^{-8}$$

$$x^2 = 6.2 \times 10^{-9}$$

$$x = 7.8740 \times 10^{-5}$$

$$= 7.9 \times 10^{-5}$$

$$\text{pH} = -\log(7.8740 \times 10^{-5})$$

$$= 4.1038$$

$$= 4.10$$

$\therefore$ The predicted pH of NaHSO_3 is 4.10 (experimental: 4.3)

Part 2: K_a Determinations and Percent Error

K_a of HSO_3^-

$$\text{pH} = \text{p}K_a + \log\left(\frac{[\text{A}^-]}{[\text{HA}]}\right) = \text{p}K_a + \log(1) = \text{p}K_a$$

$$\text{p}K_a = 6.5$$

$$K_a = 10^{-6.5}$$

$$= 3.16227 \times 10^{-7}$$

$$= 3.2 \times 10^{-7}$$

$$\% \text{ error} = \left| \frac{3.16227 \times 10^{-7} - 6.2 \times 10^{-8}}{6.2 \times 10^{-8}} \right| \times 100\%$$

$$= 409.72\%$$

$$= 4.1 \times 10^2\%$$

$\therefore$ The experimental K_a of HSO_3^- was determined to be 3.2×10^{-7} with a $4.1 \times 10^2\%$ percent error.

K_a of CH_3COOH

$$\text{p}K_a = 6.6$$

$$K_a = 10^{-6.6}$$

$$= 2.51188 \times 10^{-7}$$

$$= 2.5 \times 10^{-7}$$

$$\% \text{ error} = \left| \frac{2.51188 \times 10^{-7} - 1.8 \times 10^{-5}}{1.8 \times 10^{-5}} \right| \times 100\%$$

$$= 98.604\%$$

$$= 98.6\%$$

$\therefore$ The experimental K_a of CH_3COOH was determined to be 2.5×10^{-7} with a 98.6% percent error.

Discussion

1. The predicted pH of the NaHSO_3 solution was 4.104, while the actual experimental result was 4.3. The difference in pH can be attributed to the conditions during the experiment not being SATP, while the calculations were based on SATP conditions. At lower temperatures, the equilibrium of the reaction would shift to the left, which would lead to less dissociation of HSO_3^- and a higher pH.

2. One possible explanation for the percent error in the K_a of CH_3COOH and NaCH_3COO is the accuracy of the pH meter. Since the pH probe could only measure pH to one significant digit, the logarithmic relationship between pH and $[H^+]$ would magnify any errors in the pH measurement, leading to a large percent error in the K_a value.

Another possibility is that the concentration of the CH_3COOH and NaCH_3COO was not accurately known. Due to the age of the solutions, significant evaporation may have occurred, increasing the concentration of the solutions and thus skewing the pH values. The discrepancy between experimental and theoretical values is likely due to a combination of these factors.

3. When a strong base, such as NaOH , was added to the $\text{HSO}_3^-/\text{SO}_3^{2-}$ buffer solution listed in Table 3, the OH^- ions reacted with the H_3O^+ ions to form water ($\text{OH}^- + \text{H}_3\text{O}^+ \rightarrow 2\text{H}_2\text{O}$). This significantly reduced the concentration of H_3O^+ in the solution. According to Le Chatelier's principle, the system responded to this stress by shifting the equilibrium to the right to produce more H_3O^+ ions and restore equilibrium. This replaced most of the H^+ ions that were consumed, thus stabilizing the pH. In contrast, when NaOH was added to deionized water, there was no system in place to restore the H_3O^+ concentration, causing the pH to increase significantly as shown in Table 3.

Conclusion

In conclusion, the pH of various salt solutions was measured and predicted, and the experimental K_a values for HSO_3^- and CH_3COOH were successfully determined. These findings addressed the initial purpose of evaluating hydrolysis reactions and the buffering capacity of weak acid systems.